Pearls — Work Order for Antigravity — Thursday, October 29, 2026

Prepared by: Janine, at D'Vaughn's direction **For:** Antigravity, on the iMac **Tree:** the thinktank-os tree (`~/projects/dvaughnhouse/thinktank-os`), Pearls code at `pearls-buildout-2026-10-23/pearls/` **Branch:** `pearls-buildout` (exists locally and on origin, currently at `f46ba91`). Work, commit, and push on `pearls-buildout` only — never commit directly to `main` . **Status:** This order is self-contained for today. It supersedes the 2026-10-28 work order as the operative instruction. Phase 0 is done (see below) — today's job is Phase 1, PearlHook completion.

Where things stand (verified, not claimed)

Phase 0 is DONE and independently verified — do not redo it.

- The 2026-10-28 order was run and merged: `f46ba91` ("Phase 0 true Uniswap v4.0.0 pin @e50237c and IPoolManager struct alignment"), merged to `main` as `b72f0dc` .
- Janine independently verified it 2026-10-29: `lib/v4-core` blob hashes match the true `v4.0.0` tag byte-for-byte (`src/interfaces/IPoolManager.sol` = `85703ef79d0271f9a8816cfd7cf7a5c63e28a706` , `src/PoolManager.sol` = `c24fbc0ea6c31791617b3e15ce21d585ebd5fdf2`); `foundry.lock` reads `lib/v4-core` tag `v4.0.0` , rev `e50237c43811bd9b526eff40f26772152a42daba` ; all 5 files import from `IPoolManager.sol` ; `BUILD-OUT-REPORT.md` §§1.1/4.2 corrected and the `59d3ecf` claim retracted.
- Full suite re-run against the true pin: **59 tests passed, 0 failed, 0 skipped** (5 suites, 4 test files: `Pearls.t.sol` , `PearlHook.t.sol` , `PearlHookForkTest` , `PearlHookRouterForkTest` , `AttestationRegistryTest` — 45 unit/fuzz + 14 fork, fork tests against the public Base Sepolia RPC `https://sepolia.base.org`).
- The old 59-vs-53 count discrepancy is resolved: 59 is the iMac tree's real count, reproduced independently. The 53 figure was measured on Janine's VM canonical tree (`~/workspace/your_files/pearls/`), a different tree state. Canonical-tree sync is D'Vaughn's call, not yours.

One toolchain finding — fix it first (Task 0 below): Foundry is NOT installed on the iMac (`~/.foundry/bin/` does not exist, no `forge` on PATH). The verification above ran on a copy of the iMac tree elsewhere. You need a working `forge` where you work before Phase 1 can be verified.

Today's tasks — Phase 1: PearlHook completion (the critical path)

For each item, first assess what is already present (some pieces have tests already — e.g. a hook-miner test exists), then complete what is missing. Report per item what was already there vs what you built. Never weaken a test to make it pass.

0. Establish the toolchain on the iMac. Install Foundry (foundryup) on the iMac, verify `forge --version` and a clean `forge build` in `pearls-buildout-2026-10-23/pearls/`. If you run the suite anywhere else, say exactly where and why.

1. Pin v4-periphery. `foundry.lock` currently carries `lib/v4-periphery` at rev `3779387e5d296f39df543d23524b050f89a62917` with NO tag — it floats. Identify the release tag that rev belongs to (or the correct release tag to pin), pin it the same way v4-core is pinned (tag + rev in `foundry.lock`), and re-verify the full suite. Do not float versions. If pinning periphery surfaces API differences, fix minimally and report them.

2. Replace mirrored callback signatures with the canonical v4 interfaces. Wherever PearlHook/PearlHookCore mirrors v4 callback signatures instead of using the canonical interfaces, replace the mirrors. No behavior change beyond the interface alignment; report every file touched.

3. Complete CREATE2 salt mining so the hook address encodes its callback permissions. Verify the existing miner against the hook's actual permission flags; complete it if it is partial, and prove a mined address deploys with exactly the intended flags (and that an unmined address is rejected — the existing test names this behavior; keep it green).

4. Implement fee settlement inside the unlock context — the core of this phase. The hook must actually TAKE the fee via `PoolManager.take()` / `settle()`, not compute it and claim a delta of 0. Quote math alone is not settlement: prove on a forked pool that fees move to the right party in all four direction cases (buy/sell × exact-in/exact-out), pre- and post-gate, and that nothing is taken below the gate. This is the item the whole phase stands or falls on.

5. Run the spec §7.3 fork-test matrix against a Base Sepolia fork. 4 direction cases × gate state × guard state, on the fork (public `https://sepolia.base.org` is the default when no `BASE_SEPOLIA_RPC_URL` is set). All fork tests green, including the router-path fork tests and the guard-window cases. Report exact counts per suite.

6. Commit and push on branch `pearls-buildout`. A fix that is live but uncommitted is not done. Commit and push on the branch where it runs. Do not merge to `main` yourself — that is D'Vaughn's call.

Ground truth (verified, do not re-derive — build on it)

- Uniswap v4-core release tag **`v4.0.0` = commit `e50237c43811bd9b526eff40f26772152a42daba`** (2025-01-21). Verified by blob hash, above.

- In true v4.0.0, `SwapParams` and `ModifyLiquidityParams` live in **`src/interfaces/IPoolManager.sol`**. The file `src/types/PoolOperation.sol` **does not exist** in v4.0.0.
- `59d3ecf53afa9264a16bba0e38f4c5d2231f80bc` is a **May 2025 main-branch commit**, NOT the v4.0.0 tag. Never repeat the "v4.0.0 @ 59d3ecf" claim anywhere.
- The aggregator work (commit `958dccb`) is real and passing, verified against the true pin as part of the 59 above. The 2026-10-23 aggregator-hooks prompt (router-path matrix, guard-window characterization, quote-competitiveness benchmark, factory tier restriction, launch-event completeness, seeding shape) is a SEPARATE pending item awaiting D'Vaughn's paste — it is NOT part of today's tasks unless he says so.

Locked rules — D'Vaughn's decisions, not yours to change

If anything below looks wrong, surface it in your report. Do not remove, rename, relabel, or "fix" it on your own.

- **Fee doctrine, frozen:** the fee is configured at inception per launch, then immutable forever. No platform fee until $10M cumulative volume; above the gate, 0.30% in quote asset only (WETH/USDC); the crossing swap pays no fee; the gate is fixed in quote units at deploy (no swap-time oracle). Tier 1 Patient (locked/burned LP): $10M gate / 0.30%. Tier 3 Uncommitted (no lock): $20K gate / ~0.58% modeled — exact figure unconfirmed, do not hard-code it as locked.
- **No admin keys, no owner, no setters, no pause, no proxy, no upgrade path.** Core contracts have none and gain none.
- **Do not change fee constants, the $10M gate, or lock minimums to make tests pass.** If a design point seems wrong, flag it in the report.
- **Testnet/local only.** Never point anything at mainnet. Deployment anywhere — including Sepolia — is D'Vaughn's explicit call, not part of today's work. `script/Deploy.s.sol` is hard-gated to Base Sepolia (84532) and local Anvil (31337); do not touch that gate.
- **Canonical tree untouched.** `~/workspace/your_files/pearls/` (Janine's canonical copy) stays untouched — sync is D'Vaughn's call.
- **Naming, everywhere including code comments and variable names:** Pearls, always plural. The gold colour and Studio seal are Abyssal Gold. Never use: "sovereign", "ventures", or "institute" for Glenride. User-facing copy never says "AI" — users get help from "the Assistant". (This work order is internal; the rule binds anything user-facing you write.)
- **Honest status, always:** nothing is claimed as live, deployed, audited, or counted until it is. No fabricated testimonials, stats, or endorsements. No document may claim the hook is complete until the §7.3 fork-test matrix passes against the true pin.

Non-goals for today

No deploy. No Sepolia rehearsal (that is Phase 3, D'Vaughn's gate). No mainnet. No Phase 2 security work (Aderyn/Halmos/adversarial tests come after completion, on the next order). No changes to fee invariants, guard windows, or lock minimums. No canonical-tree edits. No landing-page work (already live at pearls.dvaughnhouse.com).

Open decisions — D'Vaughn's calls, not yours (do not decide them in code)

Pearl #0 (SPINE manual deploy vs Sepolia dummy) · routing fee · waitlist path · Plus+ portal auth · canonical-tree sync. If your work touches one, leave the current behavior intact and flag it.

Report back (required, in BUILD-OUT-REPORT.md)

- Toolchain: where `forge` now runs on the iMac (or where the suite ran, if not the iMac) + `forge --version`.
- Per Phase 1 item: what was already present vs what you built, files touched, and any API differences the periphery pin surfaced.
- Exact test counts per suite (forge, incl. fork matrix), and the settlement proof: fees taken on fork in all four directions, pre- and post-gate, zero below the gate.
- Remaining gaps, if any.

Janine independently re-verifies all of it (blob-hash vs the tag + full suite re-run) before anything is reported to D'Vaughn as sound. Report findings; do not soften them.

Definition of done for today

- `forge` works on the iMac; `forge build` clean against true v4.0.0 + pinned periphery.
- `forge test` fully green including the §7.3 fork-test matrix; settlement proven on fork, not just quote math.
- Report updated, commit pushed on `pearls-buildout`.
- Nothing deployed, nothing claimed beyond what the run proves.